

EDXSO AI Engineer Intern – Assignment 1

Automated Micro-Influencer Outreach System

Objective

Build an automated system that discovers relevant micro-influencers, filters and classifies them based on predefined criteria, enriches their profiles with useful information, and generates personalized collaboration outreach messages.

The system should demonstrate your ability to work with **APIs, web scraping/data extraction, data processing, automation, AI/LLM-based personalization, and workflow design.**

1. Influencer Discovery

Identify micro-influencers from a chosen niche such as:

- Fitness
- Fintech
- Beauty
- Fashion
- Crypto
- Parenting
- Gaming
- Lifestyle
- Technology

You may use:

- Instagram
- YouTube
- TikTok
- Hashtag searches
- Public influencer directories
- UGC marketplaces such as Collabstr, Aspire, or Grin
- Social media scraping/data extraction tools
- Creator newsletters and creator-spotlight pages

Definition

A micro-influencer should generally have **5,000–100,000 followers** or demonstrate equivalent audience engagement.

Requirement

Your system should fetch **at least 50 influencers** for the test run.

2. Filtering & Classification

Build logic to automatically filter and classify the discovered influencers.

Your system should consider:

- Category / niche
- Audience demographics
- Geography, if available
- Social media platform
- Follower count
- Engagement rate
- Brand-fit criteria
- Content relevance

Minimum Requirement

Implement at least **one complete filtering category**.

Example: Fashion & Beauty influencers.

The final output should clearly identify which influencers passed or failed the filtering criteria and why.

3. Profile Enrichment

For every shortlisted influencer, enrich the profile with the following information:

Field	Requirement
Influencer Name	Mandatory
Platform	Mandatory
Profile URL	Mandatory
Follower Count	Mandatory
Engagement Rate	Mandatory
Category / Niche	Mandatory
Content Themes	Mandatory
Contact Email	Mandatory
Instagram / YouTube / TikTok	Optional
Website	Optional
Audience Age	Optional
Audience Gender	Optional
Audience Geography	Optional

Minimum Enrichment Requirement

Each influencer record must contain:

Contact email + profile metrics + niche/content context

If an email cannot be found, clearly mark it as **"Not Found"** rather than generating or guessing one.

4. Message Personalization

The system should automatically generate **two types of personalized outreach messages** for each qualified influencer.

A. Email Collaboration Pitch

Length: **60–90 words**

The email should reference relevant personalization signals such as:

- Influencer's niche
- Content tone/style
- Recent content
- Relevant audience
- Proposed collaboration
- Value proposition

Possible collaboration angles include:

- Sponsorship
- Affiliate campaign
- UGC content creation
- Brand ambassador program
- Paid product placement
- Barter collaboration

B. Instagram DM

Length: **15–30 words**

The DM should be short, natural, and personalized based on the influencer's content and niche.

Example Personalization

Instead of:

"Hi, we would like to collaborate with you."

The system should generate something closer to:

"Hi Sarah, loved your recent skincare routine content. Your beauty-focused audience looks like a great fit for our upcoming UGC campaign."

The system should generate these messages dynamically rather than using one fixed template for every influencer.

5. Sending Layer

Build a sending layer that demonstrates how the generated outreach would be delivered.

You may implement this using:

- Gmail API
- SMTP
- Email automation tools
- Instagram/Meta APIs where permitted
- n8n
- Make
- Zapier
- Another appropriate automation workflow

Requirements

The sending layer should:

1. Select influencers with a valid contact email.
2. Retrieve their personalized email message.
3. Send or simulate sending the email.
4. Record the sending status.
5. Prevent duplicate outreach.
6. Maintain a basic outreach log.

For Instagram DMs, if direct automated sending is not technically or legally available through the chosen API, **do not bypass platform restrictions**. Instead, demonstrate the generated DM and create a simulated/manual sending workflow.

6. Suggested Workflow

Your overall system can follow this flow:

**Discovery → Data Collection → Filtering → Enrichment → AI Personalization
→ Review → Sending → Tracking**

Example:

Social Platforms / Directories



Influencer Discovery



50+ Profiles



Filtering & Classification



Profile Enrichment



AI Personalization



Email + Instagram DM



Sending Layer



Outreach Tracker

7. Expected Output

Submit the following:

A. Working System

A functional prototype demonstrating the complete workflow.

B. Influencer Dataset

A dataset containing at least **50 discovered influencers**.

Recommended format:

Name	Platform	Followers	Engagement	Niche	Email	Profile URL	Content Theme	Status
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C. Personalized Messages

Show the generated:

- Email pitch
- Instagram DM

for the shortlisted influencers.

D. Outreach Tracker

Include fields such as:

Influencer	Email	Message Generated	Sent	Date	Status
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E. Documentation

Provide a short README explaining:

- Technology stack
- APIs/tools used
- Data sources
- Discovery methodology
- Filtering logic
- Enrichment process
- AI model/prompt used
- Personalization logic
- Sending mechanism
- Limitations
- Setup instructions

8. Technical Expectations

The assignment should demonstrate practical engineering skills.

You are encouraged to use:

- Python / JavaScript
- REST APIs
- Web scraping where permitted
- Data processing
- LLM APIs
- Prompt engineering
- n8n / Make / Zapier
- Databases or structured storage
- Email APIs
- Automation workflows

The implementation should be **modular, reusable, and reasonably error-tolerant**.

9. Evaluation Criteria

Your submission will be evaluated on:

1. **Functionality** – Does the complete workflow work?
2. **Data Quality** – Are the influencer records accurate and useful?
3. **Automation** – How much of the process is automated?
4. **Filtering Logic** – Is the classification meaningful?
5. **Profile Enrichment** – Is sufficient information collected?
6. **AI Personalization** – Are messages genuinely personalized?
7. **Engineering Quality** – Is the code/workflow structured and maintainable?
8. **Error Handling** – Does the system handle missing or invalid data?
9. **Documentation** – Can another developer understand and run the project?
10. **Scalability** – Could the system be extended from 50 to 500+ influencers?

10. Submission Requirements

Submit:

- GitHub repository or project files

- README/documentation
- Working demo or screenshots/video
- Influencer dataset
- Sample personalized outreach messages
- Automation workflow, if applicable
- Setup instructions
- List of APIs/tools used

Important

Do not submit fabricated influencer information, guessed email addresses, or fake engagement metrics. Clearly identify unavailable data.

The goal is to demonstrate a **working AI-powered influencer discovery and outreach pipeline**, not merely a static dataset or presentation.